Final Materials Homework Part 2:

T = 600 # torque (in-lbf)

L = 16 # length from drive to load point (in)

h = 0.56 # new height/width

b = 0.30 # new thickness

c = 1.0 # gauge distance

E = 16.5e6 # Young's modulus, Ti-6Al-4V STA (psi)

nu = 0.34 # Poisson's ratio (typical for Ti-6Al-4V)

s_strength = 160e3 # yield strength (ksi)

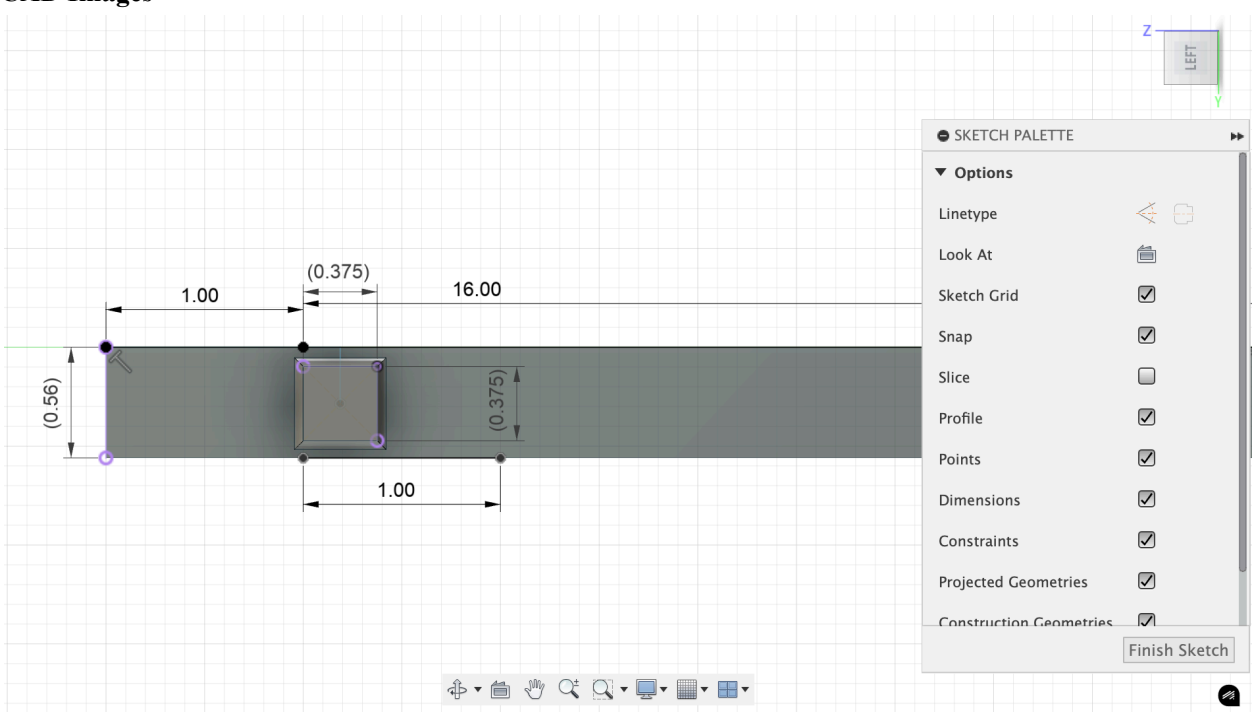
KIC = 50e3 # fracture toughness psi-sqrt(in)

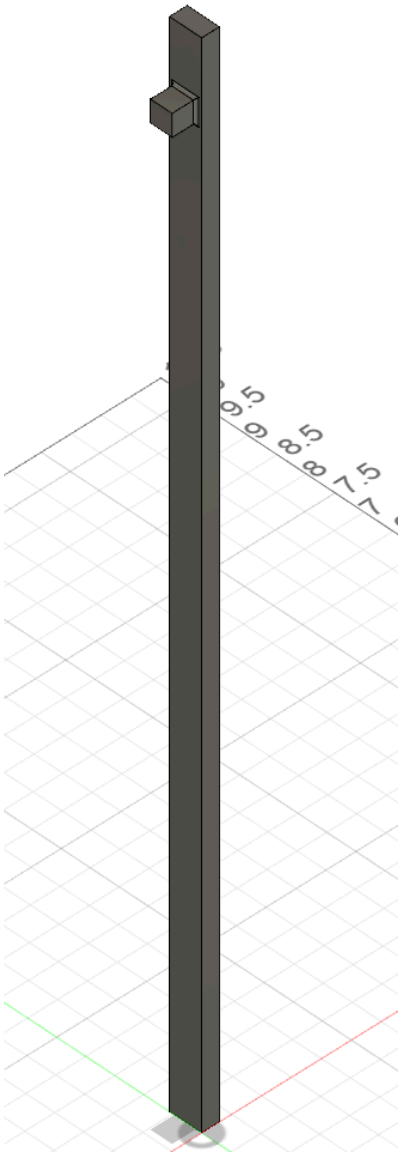
sfatigue = 90e3 # fatigue strength (psi)

a = 0.04 # assumed crack depth (in)

Load point deflection: 0.690 in
Max normal stress: 52.6 ksi
Safety factor for strength (Xo): 3.04
Safety factor for crack growth (XK): 3.46
Safety factor for fatigue (XS): 2.44
Strain at gauge: 1950 microstrain
Output: 2.07 mV/V at 600 in-lbf using half bridge

CAD Images



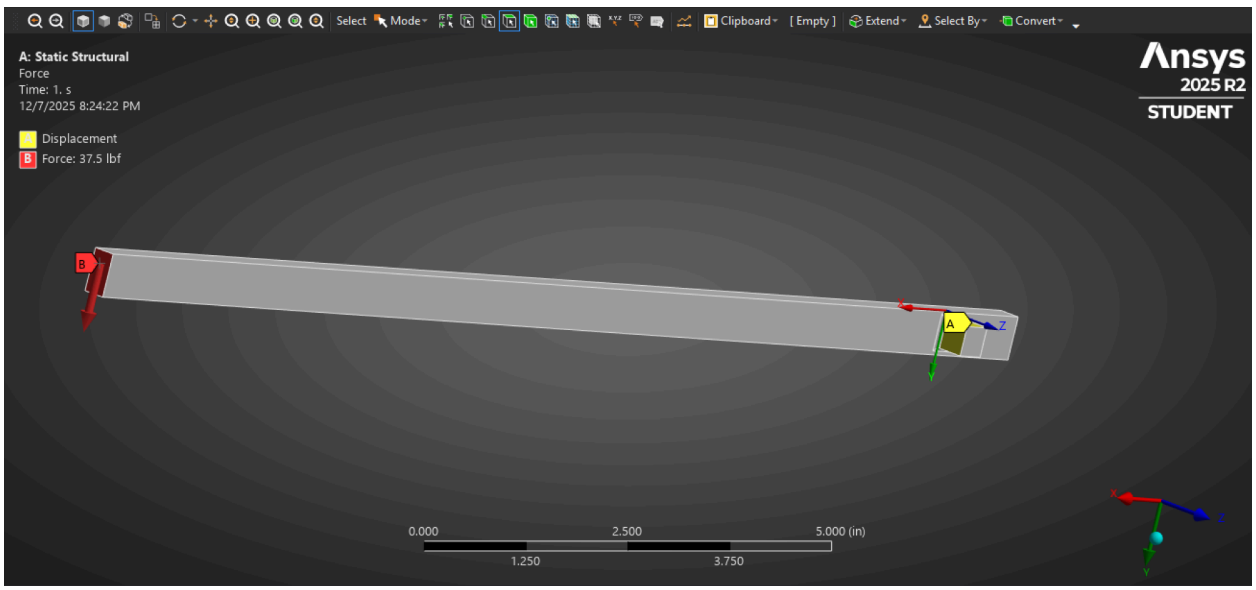

9.5
9
8.5
8
7.5
7

1. **Material Used and Relevant Mechanical Properties**

   Material: Ti-6Al-4V (Grade 5)(Solution Treated and Aged (STA))

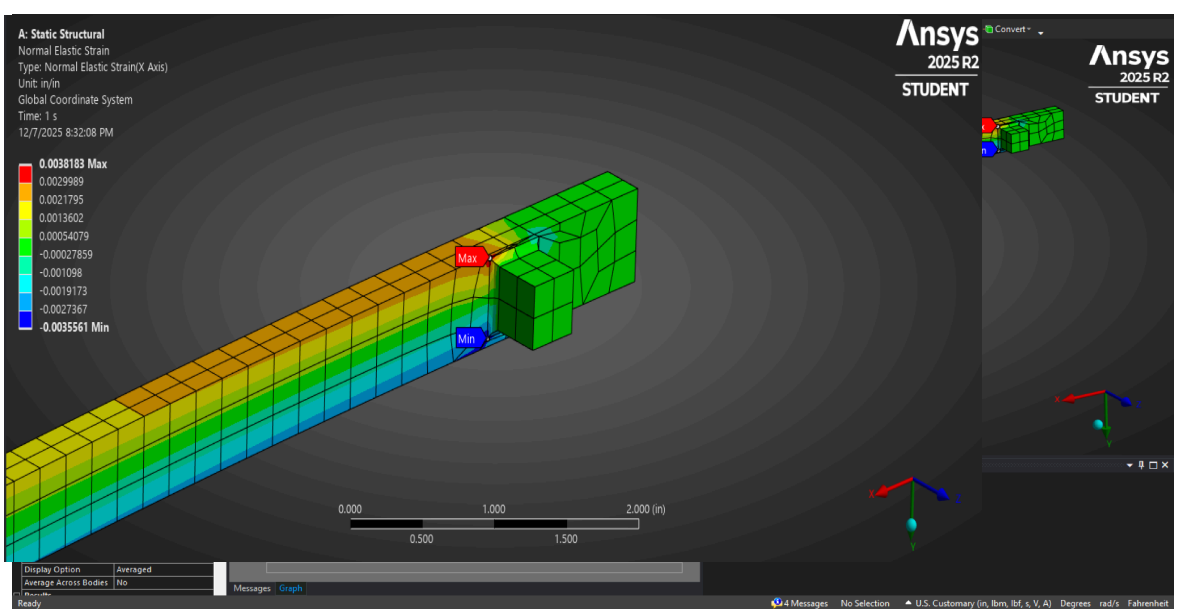
   Ti-6Al-4V STA was selected for this design because it provides a useful combination of strength and flexibility that suits the behavior required from a torque-sensing beam. The wrench needs to generate measurable strain while resisting permanent deformation and long-term fatigue damage, and titanium's mechanical properties make this easier to achieve without relying on oversized cross-sections. Compared to steel, titanium's lower modulus produces higher strain under the same bending moment, which improves the electrical output from the strain gauges. At the same time, it maintains much higher strength than aluminum, allowing the part to stay relatively compact while safely withstanding the required loading. Although titanium is more expensive, the part itself is small and the project focuses more on performance optimization than cost or scaled manufacturability. For this reason, Ti-6Al-4V provided the best balance for the specific goals of the design.

2. **FEM Loads and Boundary Conditions**

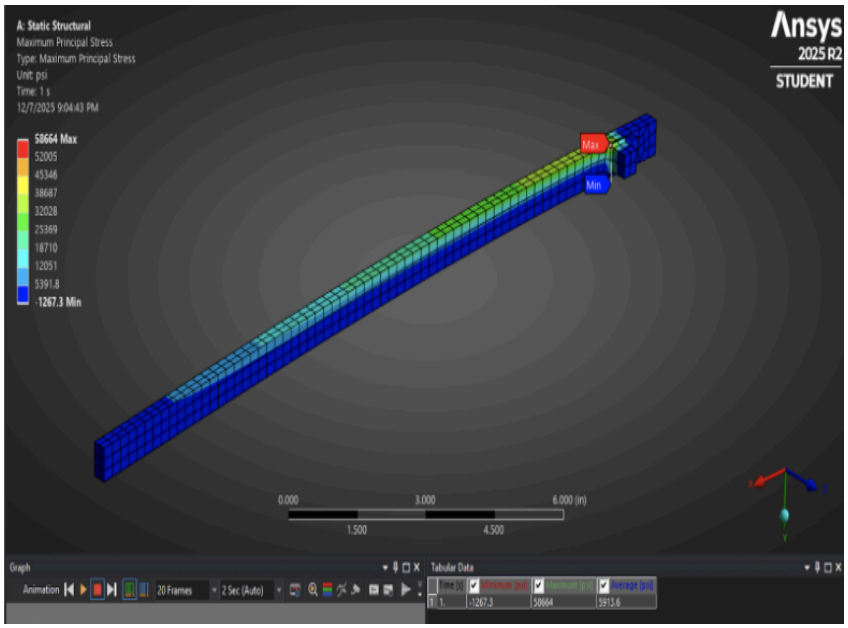


   In the FEM model, Point A (yellow) is fully fixed to represent the wrench being anchored at the drive. Point B (red) applies a 37.5-lbf load in the Y-direction, generating the required 600 in-lbf torque. This setup captures the primary bending condition the wrench experiences during use.

3. **Normal strain contours (in the strain gauge direction) from FEM**

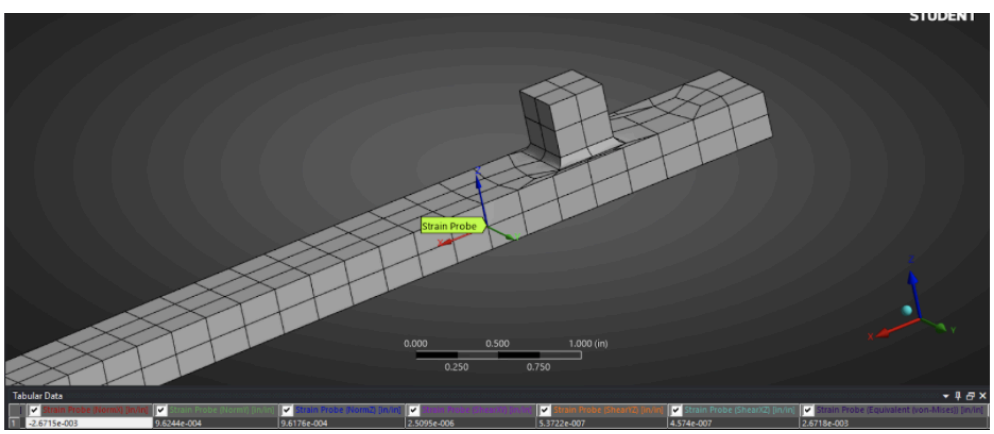


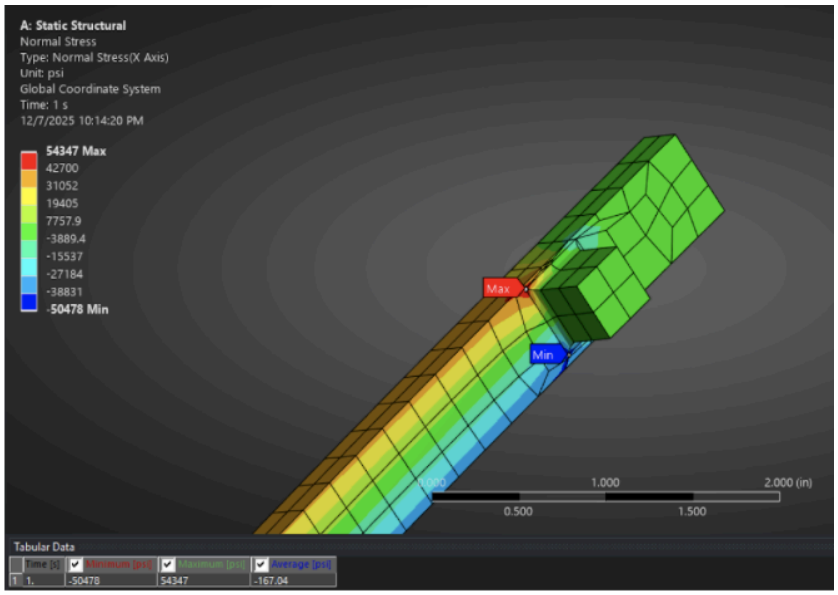
4. **Contour plot of maximum principal stress from FEM**

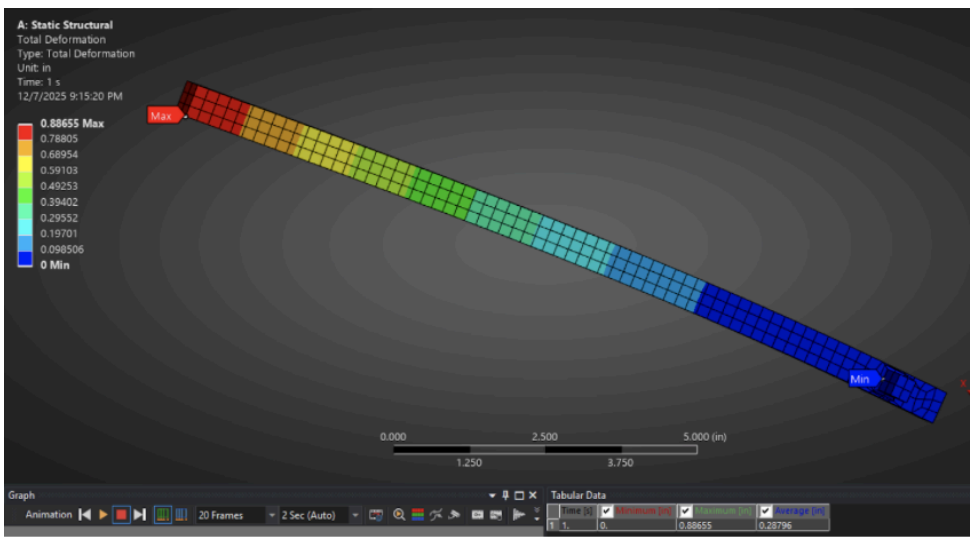


5. **Summarize results from FEM calculation showing maximum normal stress (anywhere), load point deflection, strains at the strain gauge locations**

The FEM shows that the wrench reaches a maximum normal stress of 54,347 psi, occurring at the fixed drive end where bending is most severe. The deformation results show a maximum load-point deflection of 0.88655 in, with an average deflection along the handle of 0.28796 in. At the strain-gauge location, the strain probes report a maximum principal strain of 9.6254×10^{-4} in (approx. 962 microstrain), while the minimum principal strain reaches -2.6715×10^{-3} in on the opposite face, confirming the expected tension–compression behavior under bending.







6. **Using the strain predicted by the FEM model, the torque wrench produces an output of** 0.96254 mV/V
   This sensitivity is very close to what commercial 3/8-inch torque sensors deliver, which typically fall around 1 mV/V. That similarity is important because it shows that the simulated wrench behaves like real tools already used in industry.
7. **Strain gauge selected (give type and dimensions). Note that design must physically have enough space to bond the gauges**
   The strain gauge chosen for this design is the Vishay Micro-Measurements CEA-06-125UT-350, which has a resistance of 350 Ω, a gauge factor of 2.125, and an active grid size of 0.125 in × 0.110 in. Two gauges are installed on opposite sides of the handle at a location 1.0 inch from the drive centerline so that one gauge experiences tension while the other experiences compression during loading. With the updated handle height of 0.56 inches, the available bonding surface provides ample

room for proper installation, ensuring the gauges have enough space for adhesion and strain transfer without interfering with edges or transitions in the geometry.